

Abstract

“TrackIT is a web-based ERP solution developed to streamline and automate the attendance tracking and task management process in educational institutions, particularly within the IT department. The project addresses the limitations of traditional methods such as paper registers and Excel sheets by integrating modern web technologies and Google APIs. By facilitating accurate attendance recording, real-time notifications, and centralized data management, TrackIT significantly enhances operational efficiency and communication between faculty and students.”

Introduction

“In many academic institutions, the conventional process of tracking student attendance and assigning academic tasks is highly manual and time-consuming. Faculty members often face challenges maintaining accurate records, issuing timely reminders, and managing lecture schedules. The TrackIT system was conceptualized to resolve these inefficiencies. It provides a centralized digital platform where attendance data can be submitted via Google Forms, stored securely in a database, and made instantly available to stakeholders. In addition, the system features modules for academic task management and reminders, improving coordination and accountability among students and faculty.”

Objectives

- **Automate attendance tracking:** Eliminate manual errors using Google Forms integration.
- **Enable real-time access:** Allow students and faculty to instantly view attendance.
- **Send automated notifications:** Alert users about lectures and low attendance via email.
- **Generate attendance reports:** Export records in PDF, Excel, or CSV formats.
- **Provide task management tools:** Let users create to-do lists with reminders.
- **Improve academic communication:** Share notes and notices through the system.
- **Ensure secure access:** Use role-based logins to protect sensitive data.

Methodology

“TrackIT uses a web-based architecture with a user-friendly frontend built using HTML, CSS, JavaScript, and Bootstrap. The backend is developed in Java (Servlets/JSP or Spring Boot), connected to a MySQL database. Attendance is collected via Google Forms and stored automatically. Email notifications are sent using the JavaMail API.

The system follows a role-based structure, starting with login/signup. Students can view and request attendance, and access shared notes. Faculty are able to mark attendance, manage timetables, send alerts, and upload study materials. Admins validate attendance records, verify schedules, and monitor the system’s overall functionality.”

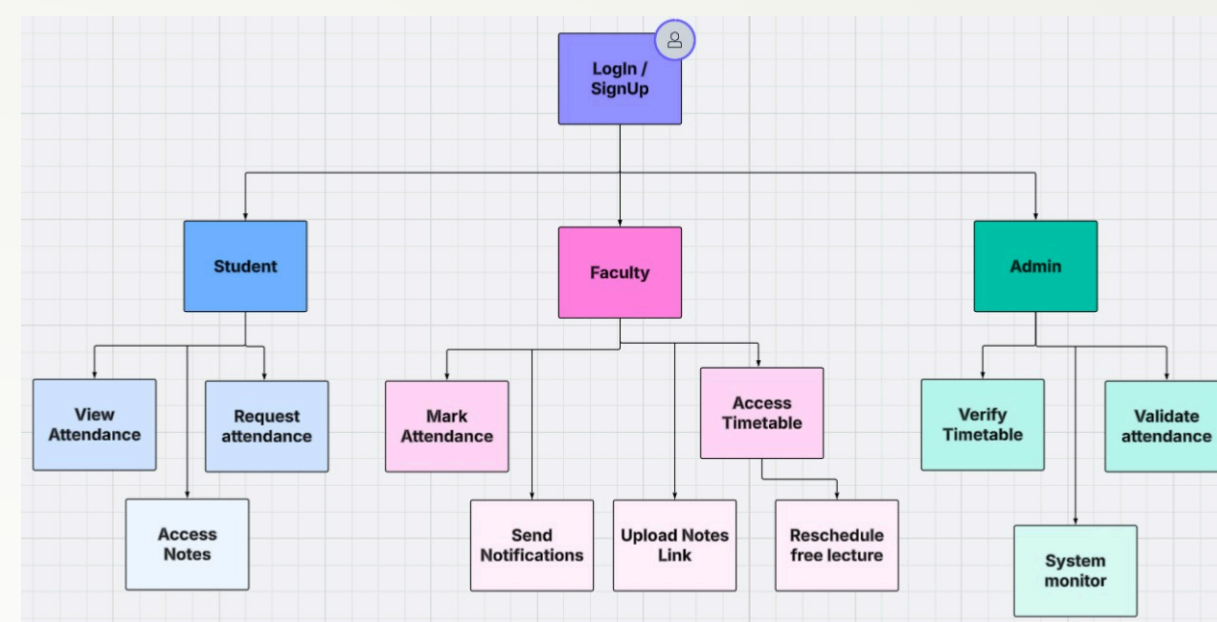


Fig.1 Flow Diagram

Results

“The implementation of TrackIT resulted in a notable improvement in the efficiency and accuracy of attendance tracking. During testing, the time required to manage attendance reduced significantly—by nearly 80%—compared to traditional methods. Real-time email alerts helped students stay informed about their attendance status, and faculty members were able to automate repetitive administrative tasks. The addition of task reminders and note sharing improved academic engagement and communication. Furthermore, the system’s role-based authentication ensured secure access to sensitive data, maintaining the integrity of academic records.”

Conclusion

“TrackIT presents a forward-thinking solution to the long-standing challenges faced by academic institutions in managing attendance and task coordination. By harnessing the power of web-based technologies and integrating features like reminders, data analytics, and digital communication tools, the platform not only reduces administrative burden but also fosters better academic management. It is flexible enough to be adapted across various educational and training environments and holds strong potential for future scalability.”

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